Who Leaves Teaching, and Why?

The Anatomy of Teacher Attrition in the United States, 1997–2024

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Figures and appendix material   July 2026

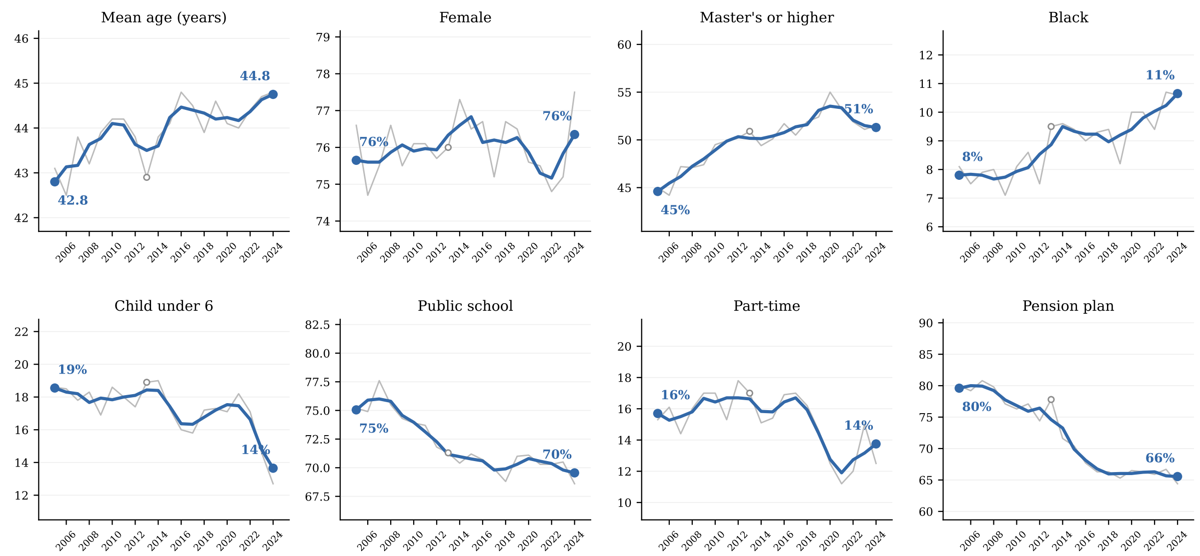

Figure 1. Composition of the Teaching Workforce, 2005–2024

Notes: College-graduate teachers aged 18+ whose longest job last year was K–12 teaching; ASEC supplement weights. Thin gray lines are annual values; blue lines are centered 3-year averages, with endpoint values printed. "Public school," "part-time," and "pension plan" refer to the longest job held last year; the open circle marks 2013, a year with a reduced sample due to the ASEC redesign. *Source:* Author's calculations from March CPS supplements, survey years 2006–2025.

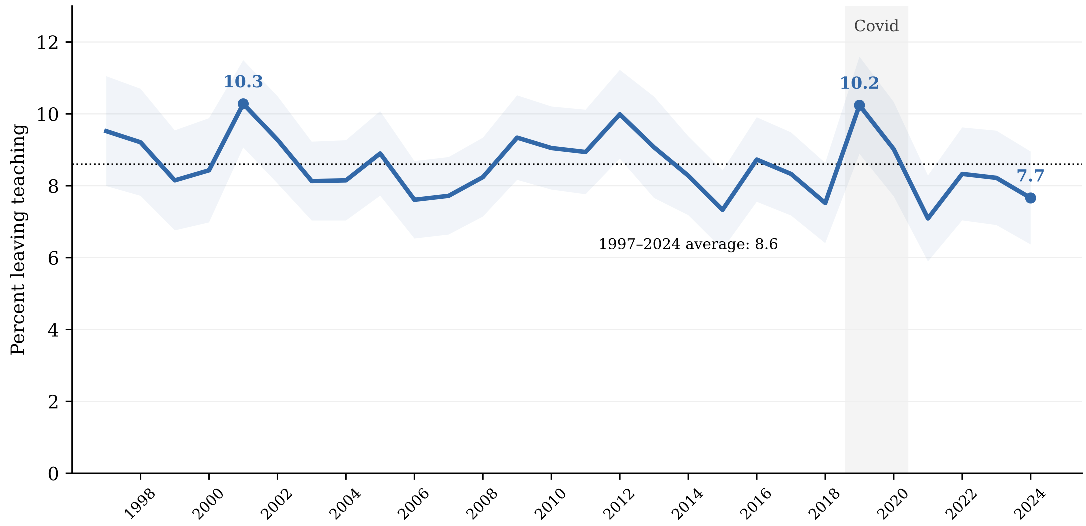

Figure 2. Leaving Teaching, 1997–2024

Notes: Share of college-graduate teachers (longest job last calendar year) not employed in a teaching occupation the following March, by route-based definition; ASEC weights. Shaded band: 95 percent confidence interval using a design factor of 1.5. Dotted line: the 1997–2024 mean (8.6). The gray band marks the Covid supplements. *Source:* Author's calculations from March CPS supplements, survey years 1998–2025.

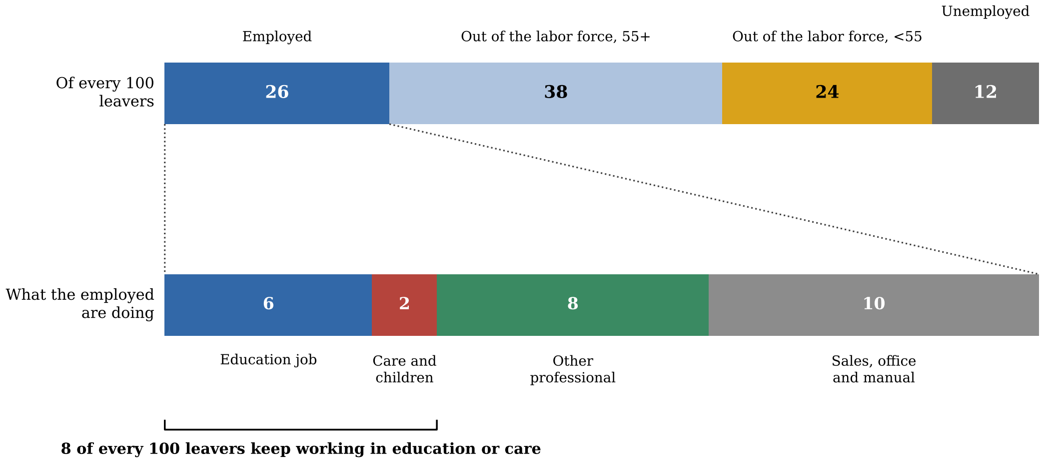

Figure 3. Destinations of Teachers Who Leave the Profession

Notes: College-graduate leavers pooled over calendar 2015–2024, ASEC weights. Top bar: labor force status the March after the teaching year; “out of the labor force” is split at age 55. Bottom bar: the occupational destination of the 26 employed leavers, rescaled to sum to their total. *Source:* Author’s calculations from March CPS supplements.

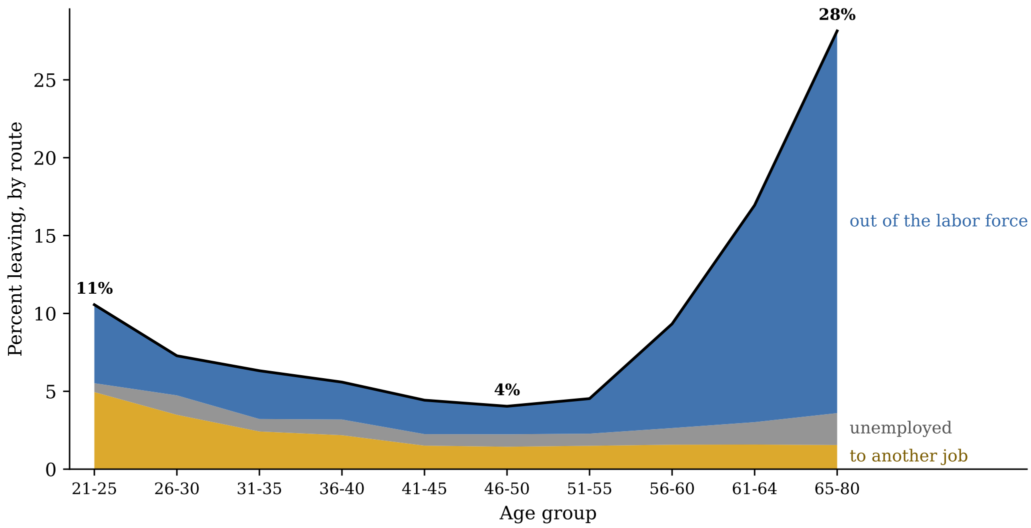

Figure 4. Leaving Teaching by Age and Route

Notes: College-graduate teachers pooled over calendar 2015–2024, ASEC weights; ten age groups. The black line is the total leaving rate; stacked areas decompose it into exits to another job (gold), unemployment (gray), and out of the labor force (blue). *Source:* Author's calculations from March CPS supplements.

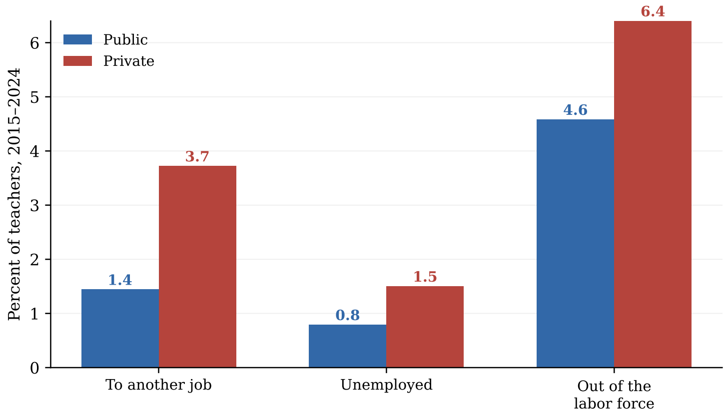

Figure 5. Exit Routes by School Sector

Notes: Route-specific annual exit rates of college-graduate teachers by sector of the last-year job (class-of-worker: government vs. private), pooled 2015–2024, ASEC weights. A private-school teacher who moves to a public school—or vice versa—remains a teacher and is not a leaver here. *Source:* Author's calculations from March CPS supplements.

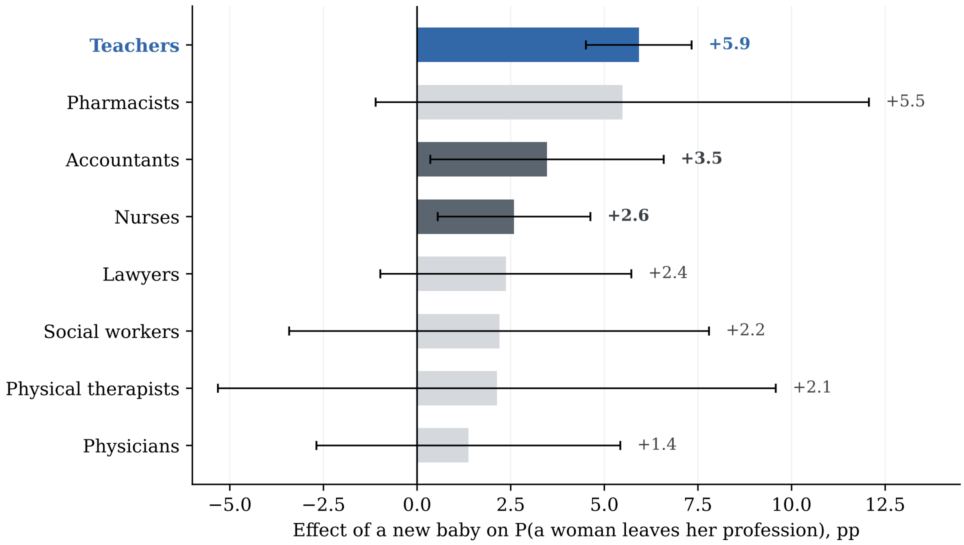

Figure 6. The Effect of a New Baby on Leaving, by Profession

Notes: Average marginal effect of a new baby (own child aged 0 in the household) on the probability that a woman leaves each profession, from profession-specific probits with age, age squared, marital status, education, and year effects; whiskers are 95 percent confidence intervals. Solid bars are significant at 5 percent; light bars are not. Samples pool 2010–2024. *Source:* Author's calculations from March CPS supplements.

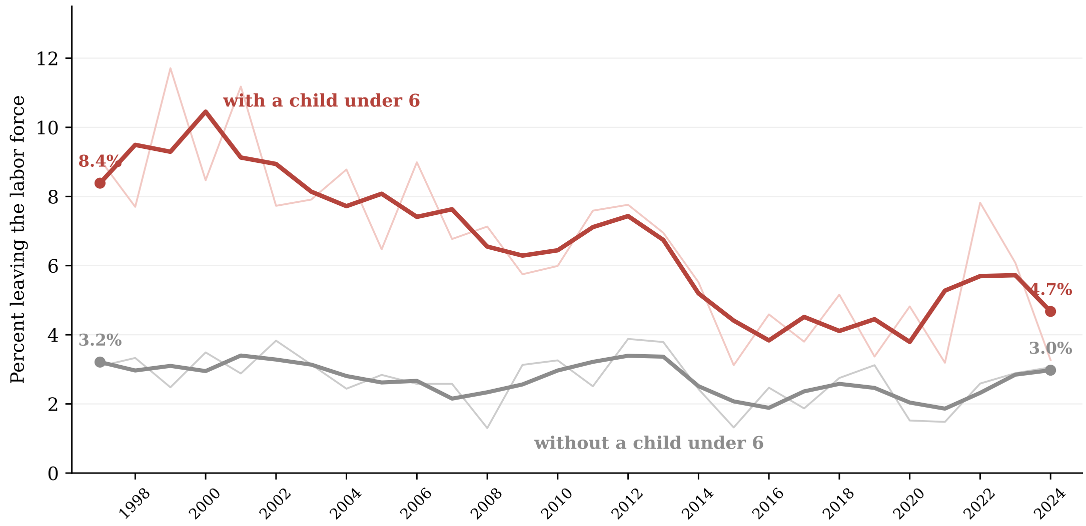

Figure 7. Labor Force Exit of Women Teachers, With and Without Young Children

Notes: Women teachers aged 22–45, college graduates; annual probability of leaving the labor force (thin lines) and centered 3-year averages (bold), separately for those with and without an own child under 6 at home. Endpoint values printed. *Source:* Author's calculations from March CPS supplements, survey years 1998–2025.

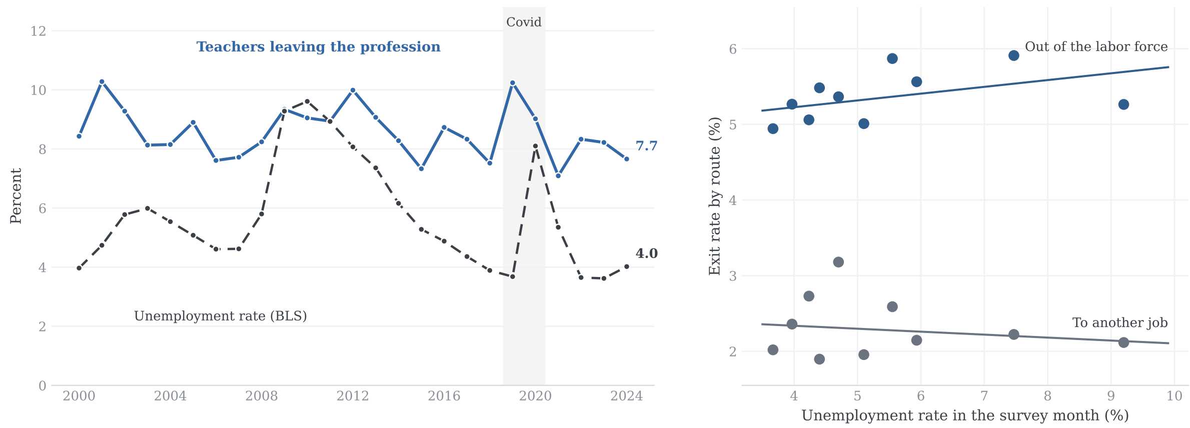

Figure 8. Leaving Teaching and the Unemployment Rate, 1997–2024

Notes: Left: the leaving series of Figure 2 (annual, from calendar 2000) against the BLS national unemployment rate (annual average). Right: binned scatter (deciles of the unemployment rate in the survey month, March of $t+1$, when the exit is observed) of the two labor-market routes, with OLS fits on the underlying annual data; correlations are -0.16 (to another job) and $+0.29$ (out of the labor force). Unemployment: BLS via FRED (`UNRATE`). *Source:* Author's calculations from March CPS supplements.

Figure 9. Leaving and State Unemployment: Two Examples

Notes: Annual state leaving rates of college-graduate teachers (dots; one per year, 1997–2024) against the state unemployment rate in the survey month (BLS Local Area Unemployment Statistics), with OLS fits. North Carolina and New Jersey are shown as examples of the two signs; the full 51-state distribution of slopes is in Figure 14. State of residence is recovered from the household records of the raw files. *Source:* Author's calculations from March CPS supplements.

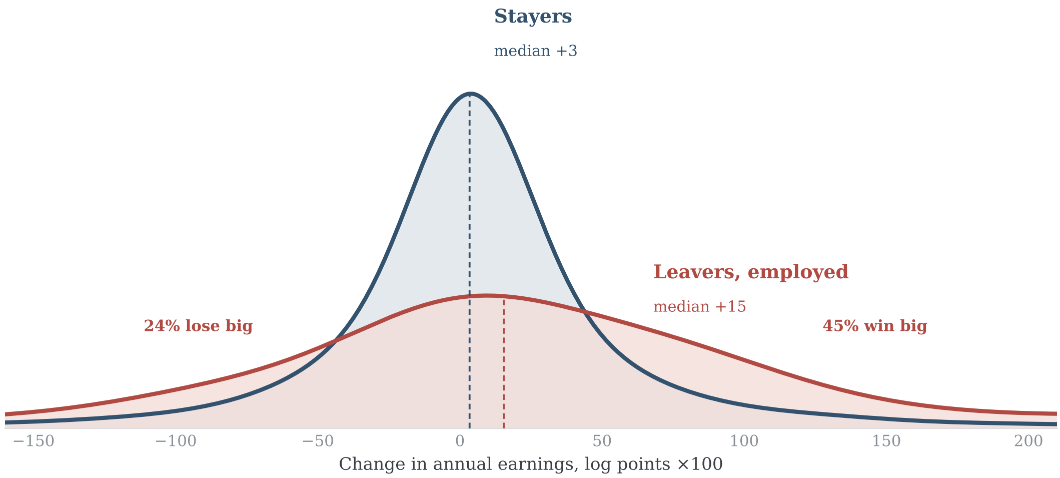

Figure 10. The Distribution of Earnings Changes: Stayers and Leavers

Notes: Kernel densities of the person-level change in annual wage and salary earnings (log points $\times 100$) between the teaching year and the next, for teachers linked across consecutive supplements (rotation-group links, survey years 2011–2025; identity validated on sex and age). Navy: stayers (12,195 links; median $+3$). Red: leavers employed the following year (225 links; median $+15$). "Lose big"/"win big" denote changes beyond ± 30 log points. The sample conditions on positive earnings in both years; 47 percent of all leavers report zero wage and salary earnings the following year. Bandwidth 0.25. *Source:* Author's calculations from March CPS supplements.

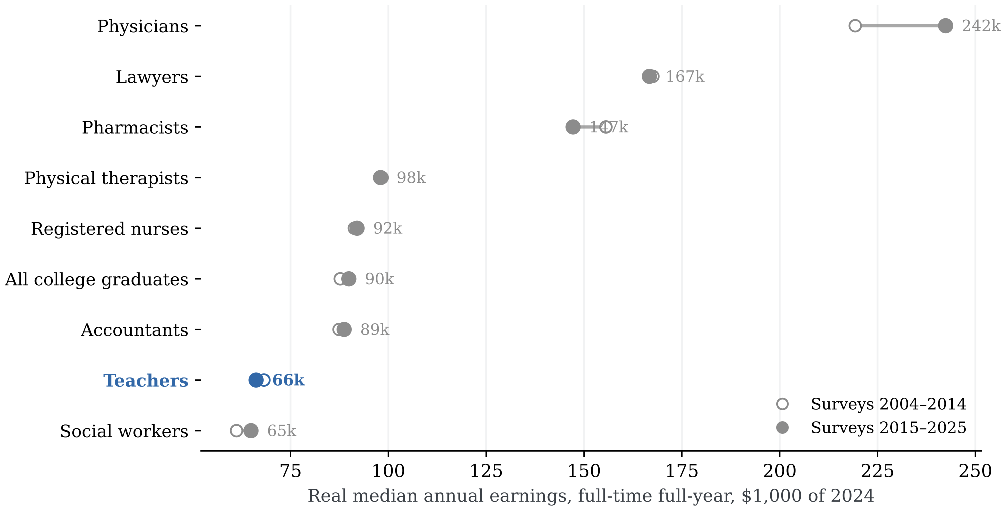

Figure 11. Real Median Earnings by Profession, 2004–2014 and 2015–2025

Notes: Weighted median annual wage and salary earnings of full-time, full-year college graduates aged 25–60, in each profession and overall, deflated to 2024 dollars with the CPI-U; the two windows pool survey years 2004–2014 and 2015–2025. Filled dots (with values): recent window. Occupation codes harmonized across classification regimes. *Source:* Author's calculations from March CPS supplements; CPI: BLS via FRED.

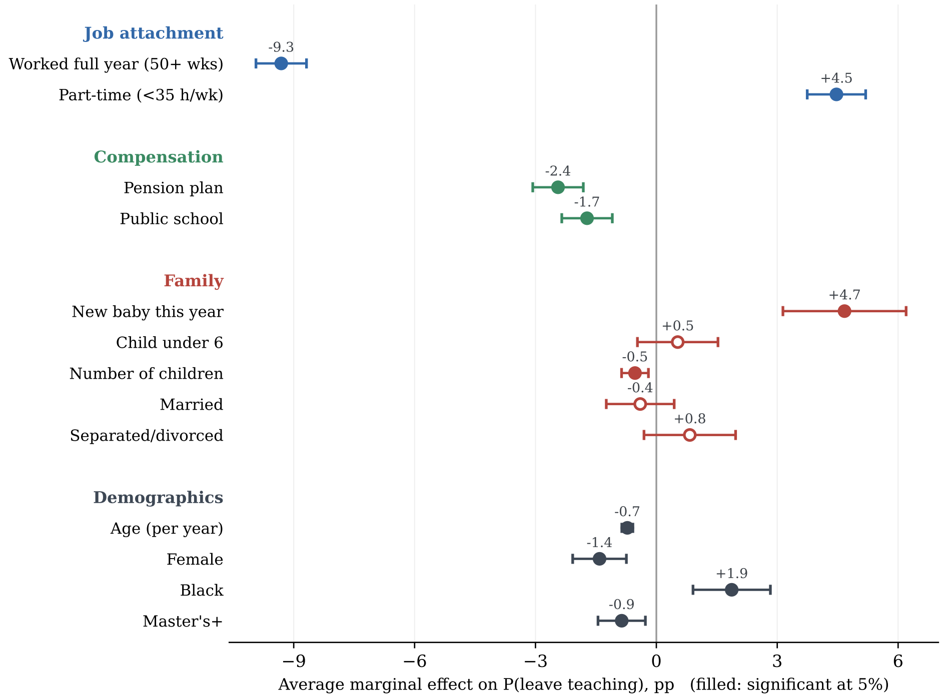

Figure 12. Average Marginal Effects on the Probability of Leaving Teaching

Notes: Average marginal effects (percentage points) on the probability of leaving teaching, from a probit on college-graduate teachers pooled over calendar 2015–2024 with year effects; whiskers are 95 percent confidence intervals; filled markers are significant at 5 percent. "Worked full year" is 50+ weeks at the last-year job; "part-time" is under 35 weekly hours. n = 28,900. *Source:* Author's calculations from March CPS supplements.

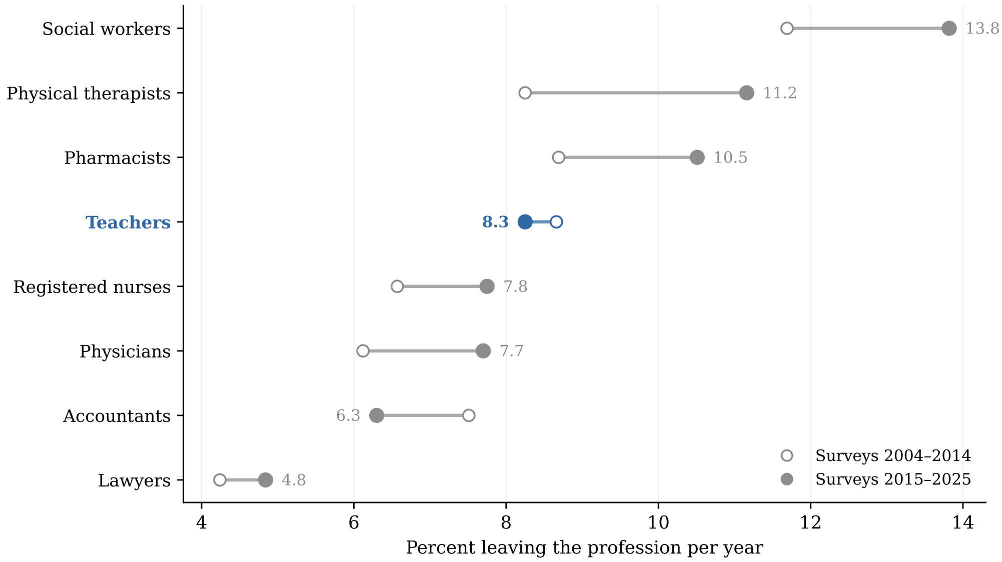

Figure 13. Rates of Leaving by Profession, 2004–2014 and 2015–2025

Notes: Annual rate of leaving each profession (same route-based definition as for teachers, college graduates, ASEC weights), pooled over survey years 2004–2014 (open) and 2015–2025 (filled, with values). Occupation codes harmonized across the 2002, 2010, and 2018 Census classification regimes. *Source:* Author's calculations from March CPS supplements.

Appendix Figures

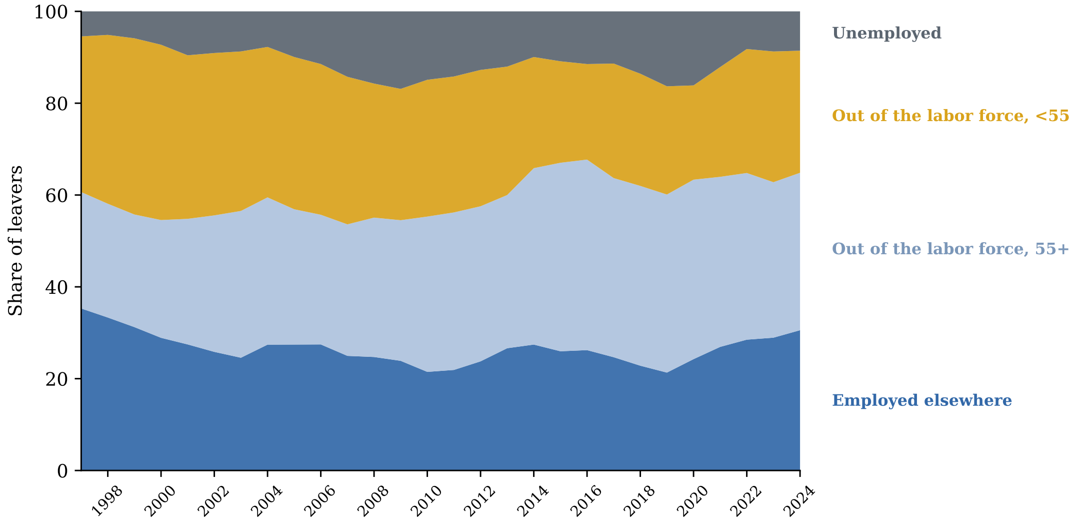

Figure 14. Route Composition of Leavers, 1997–2024

Notes: Shares of college-graduate leavers by route (3-year averages, renormalized to 100). The retirement-age share rises from roughly a quarter to nearly forty percent; the young out-of-labor-force (family) share falls symmetrically. *Source:* Author's calculations from March CPS supplements.

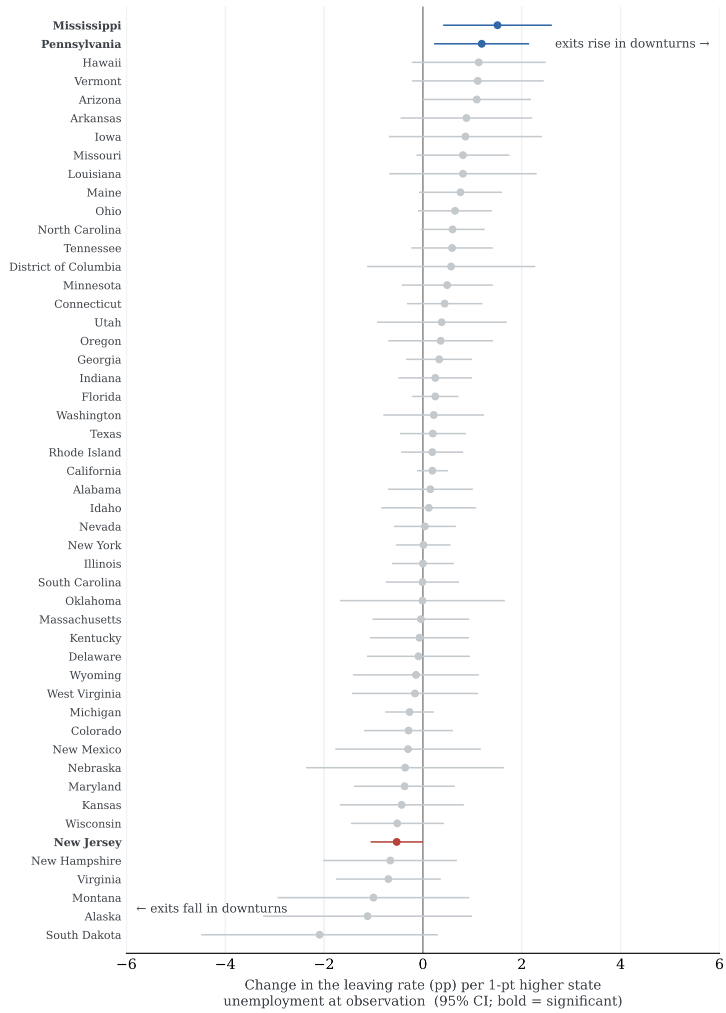

Figure 15. State-Level Cyclicality of Leaving, All 51 Jurisdictions

Notes: Slope of a weighted linear probability model of leaving on the state unemployment rate in the survey month, estimated separately by state on all years 1997–2024; 95 percent intervals, heteroskedasticity-robust. Bold: significant at 5 percent. *Source:* Author's calculations from March CPS supplements and BLS LAUS.

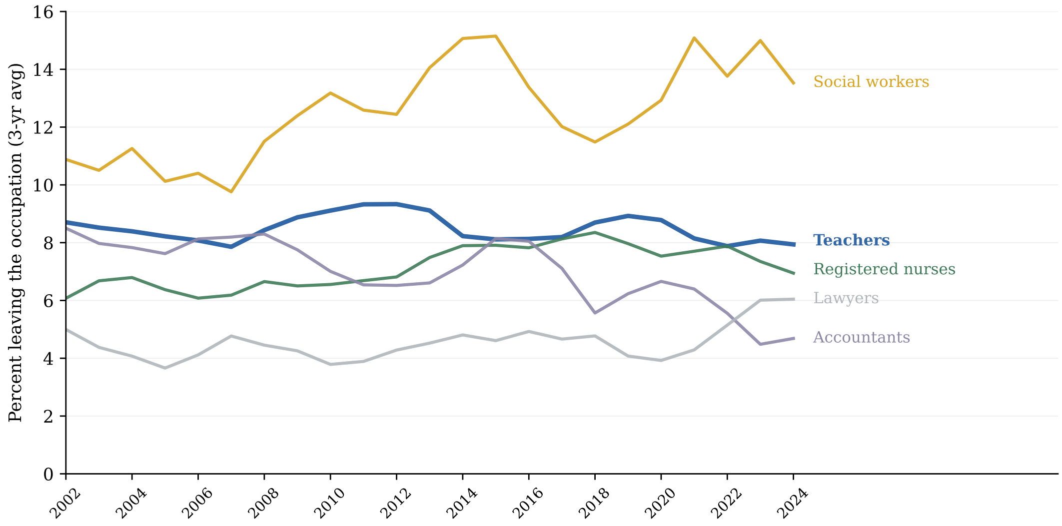

Figure 16. Leaving Rates in Five Professions, Annual Series

Notes: Three-year averages of the annual occupational leaving rate, college graduates, identical definitions across professions; the window figures pool these series. *Source:* Author's calculations from March CPS supplements, survey years 2003–2025.

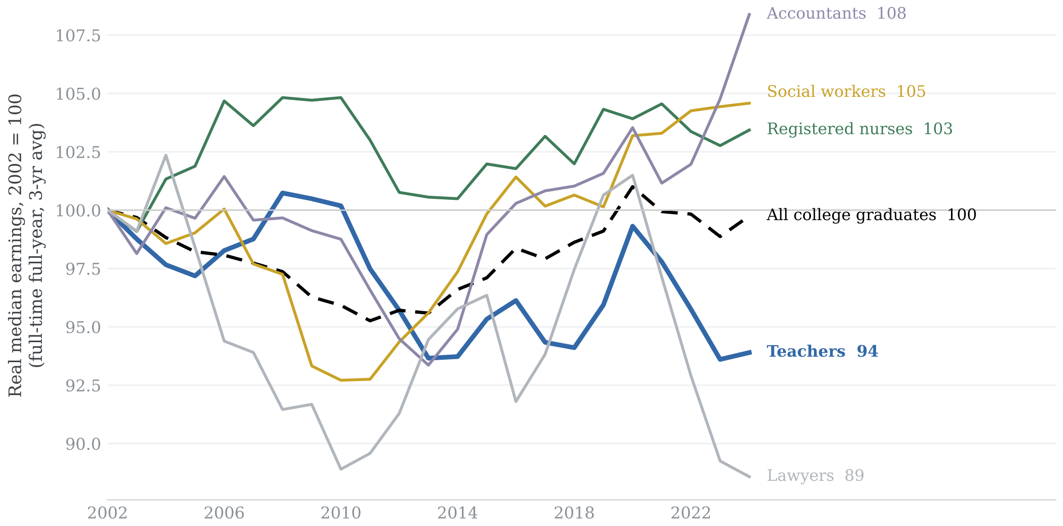

Figure 17. Real Median Earnings by Profession, Indexed to 2002

Notes: Real median annual earnings of full-time full-year college graduates aged 25–60 by profession, CPI-U deflated, indexed to 2002 = 100, 3-year averages; endpoint values printed. *Source:* Author’s calculations from March CPS supplements.

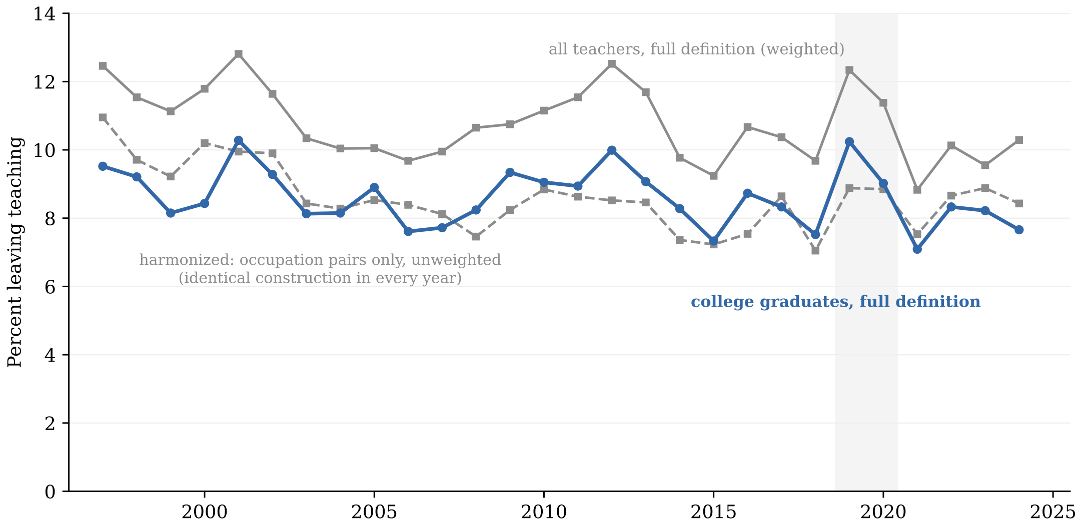

Figure 18. The Leaving Series under Alternative Definitions

Notes: The baseline route-based college-graduate series alongside the all-education series and the harmonized occupation-pair-only construction. Levels differ by universe; profiles coincide. *Source:* Author's calculations from March CPS supplements.

Appendix Tables

Table 1. March Retrospective versus Linked-Panel Classification

	March retrospective classification			
Linked-panel classification	Not a teacher	Stayed	Left	Total
Stayer (teaches all 4 post months)	198	7,911	67	8,176
Leaver (teaches in none)	1,018	10	257	1,285
Total	1,216	7,921	324	9,461

Notes: Pooled linked sample, survey years 2011–2024. Rows: 4-month within-panel classification; columns: March retrospective classification. *Source:* Author's calculations from CPS basic monthly files and March supplements.

Table 2. Leaving Rates and Routes: 1992–2001 versus 2015–2024

	Harris & Adams, 1992–2001				This paper, 2015–2024			
	All	Switch	Unemp.	Out LF	All	Switch	Unemp.	Out LF
Teachers	7.73	2.59	0.61	4.53	8.25	2.12	1.00	5.12
Nurses	6.09	1.68	0.86	3.54	7.66	3.31	0.80	3.54
Social workers	14.94	10.87	1.34	2.73	13.55	9.45	1.22	2.89
Accountants	8.01	4.10	1.55	2.36	6.30	1.79	1.18	3.32

Notes: Left block: Harris and Adams (2007), Table 1, March CPS 1992–2001. Right block: author's calculations, identical definitions, March CPS 2015–2024 (survey years 2016–2025). All rates in percent per year. *Source:* Harris and Adams (2007) and author's calculations from March CPS supplements.

Table 3. Record Counts by Source

Source	Survey years	Persons	Teacher-years	Teacher-years (BA+, 18+)
Machine-readable ASEC files	2011–2025	2,585,797	54,620	45,296
Fixed-width files, official layouts	1998–2010	2,423,332	49,925	40,201
Total	1998–2025	5,009,129	104,545	85,497

Notes: Teacher-years: longest job last year in a K–12 teaching occupation. The baseline analysis universe is college-graduate teachers aged 18+. Calendar-year coverage is 1997–2024. *Source:* Author's calculations from March CPS supplements.

Table 4. Descriptive Statistics: Teachers and Other College Graduates, 2015–2024

	Teachers, women	Teachers, men	All teachers	Other college workers
Demographics				
Age (years)	44.2 (12.8)	44.9 (13.3)	44.3 (12.9)	44.1 (13.6)
Female	100.0	0.0	76.0	48.7
Black	9.6	9.5	9.6	9.6
Married	66.6	67.7	66.9	62.6
Child under 6	16.1	17.2	16.4	15.0
Number of children	0.8 (1.1)	0.8 (1.1)	0.8 (1.1)	0.6 (1.0)
Education				
Master's degree or higher	51.3	53.6	51.9	35.4
Job held last year				
Government employer	70.3	70.4	70.3	15.5
Part-time (<35 h/wk)	15.2	11.4	14.3	13.5
Full year (50+ weeks)	71.1	74.8	72.0	85.0
Pension plan at work	66.4	67.3	66.6	46.1
Wage earnings, median ($)	50,000	58,000	52,000	70,000
Transitions				
Leaves teaching within a year	7.9	9.1	8.2	–
Person-years (unweighted N)	22,097	6,785	28,882	277,364

Notes: Weighted means; standard deviations in parentheses for continuous variables. Wage earnings are medians, current dollars, conditional on positive earnings. "Leaves teaching" is undefined for non-teachers. *Source:* Author's calculations from March CPS supplements.

Table 5. Robustness of the Attrition Model

	(1) Probit	(2) LPM, wtd.	(3) Logit	(4) LPM, FE	(5) Probit, full	(6) LPM, pair
Age (years)	-0.80***	-1.40***	-0.68***	-1.38***	-0.83***	-1.28***
	(0.08)	(0.13)	(0.07)	(0.13)	(0.05)	(0.13)
$Age^2/100$	0.96***	1.71***	0.84***	1.69***	0.99***	1.53***
	(0.08)	(0.15)	(0.07)	(0.15)	(0.05)	(0.14)
Female	-1.42***	-1.68***	-1.35***	-1.61***	-1.34***	-1.49***
	(0.34)	(0.45)	(0.34)	(0.45)	(0.21)	(0.42)
Black	1.86***	2.22***	1.77***	2.39***	2.10***	1.44**
	(0.49)	(0.69)	(0.49)	(0.72)	(0.31)	(0.63)
Married	-0.23	0.49	-0.23	0.47	-0.32	0.46
	(0.47)	(0.58)	(0.49)	(0.59)	(0.29)	(0.54)
Separated/divorced/widowed	0.96	1.78**	0.97	1.73**	0.29	1.56**
	(0.59)	(0.78)	(0.59)	(0.78)	(0.36)	(0.73)
Master's degree or higher	-0.84***	-0.50	-0.85***	-0.45	-0.58***	-0.59*
	(0.30)	(0.38)	(0.30)	(0.40)	(0.18)	(0.35)
Public school	-1.74***	-1.56***	-1.64***	-1.59***	-1.28***	-1.24***
	(0.32)	(0.47)	(0.32)	(0.48)	(0.20)	(0.44)
Pension plan at work	-2.46***	-2.57***	-2.47***	-2.56***	-3.11***	-2.08***
	(0.32)	(0.45)	(0.32)	(0.45)	(0.20)	(0.42)
Part-time (<35 h/wk)	4.37***	8.25***	3.95***	8.22***	4.62***	6.82***
	(0.37)	(0.85)	(0.35)	(0.85)	(0.22)	(0.79)
Worked full year (50+ wks)	-9.26***	-12.47***	-9.66***	-12.55***	-10.32***	-9.77***
	(0.32)	(0.54)	(0.33)	(0.54)	(0.19)	(0.49)
Number of children	-0.47***	-0.38*	-0.65***	-0.37*	-0.72***	-0.33*
	(0.17)	(0.20)	(0.18)	(0.20)	(0.11)	(0.18)
Child under 6	0.45	0.50	0.52	0.54	0.62**	0.67
	(0.51)	(0.50)	(0.54)	(0.51)	(0.30)	(0.47)
New baby this year	4.61***	5.42***	4.85***	5.44***	4.76***	4.78***
	(0.78)	(1.23)	(0.77)	(1.23)	(0.43)	(1.17)
Observations	28,900	28,900	28,900	28,900	85,430	28,900

Notes: The dependent variable is an indicator for leaving teaching, scaled so that coefficients are percentage points. Column 1 reports baseline probit average marginal effects, calendar 2015–2024, with year effects, unweighted. Column 2: linear probability model, ASEC weights, heteroskedasticity-robust standard errors. Column 3: logit average marginal effects. Column 4: weighted linear probability model with state and year effects. Column 5: probit on the full 1997–2024 window. Column 6: weighted linear probability model with the occupation-pair definition of leaving. Standard errors in parentheses.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Table 6. The Attrition Model in Subsamples

	Women	Men	Public	Private	Under 45	45+
Age (years)	-1.35***	-1.41***	-1.38***	-1.44***	-0.00	-1.31**
	(0.15)	(0.28)	(0.16)	(0.25)	(0.51)	(0.59)
$Age^2/100$	1.66***	1.69***	1.77***	1.61***	-0.22	1.65***
	(0.17)	(0.30)	(0.18)	(0.26)	(0.73)	(0.51)
Female	–	–	-0.71	-4.06***	-1.20**	-2.05***
			(0.48)	(1.03)	(0.59)	(0.70)
Black	2.45***	1.20	2.01***	2.41	3.19***	0.74
	(0.77)	(1.48)	(0.75)	(1.47)	(1.00)	(0.91)
Married	0.18	1.72	0.81	-0.07	0.62	-0.29
	(0.66)	(1.23)	(0.64)	(1.24)	(0.70)	(1.05)
Separated/divorced/widowed	1.43	2.94*	2.25**	0.46	1.34	1.07
	(0.88)	(1.76)	(0.88)	(1.58)	(1.05)	(1.21)
Master's degree or higher	-0.28	-1.51*	-0.53	-0.48	-1.37***	0.13
	(0.43)	(0.83)	(0.41)	(0.81)	(0.48)	(0.59)
Public school	-1.12**	-3.11***	–	–	-3.53***	0.61
	(0.53)	(1.04)			(0.60)	(0.74)
Pension plan at work	-1.95***	-4.54***	-2.99***	-2.03**	-2.50***	-2.74***
	(0.50)	(1.01)	(0.54)	(0.80)	(0.56)	(0.71)
Part-time (<35 h/wk)	7.86***	10.63***	10.96***	5.37***	7.96***	7.68***
	(0.93)	(2.07)	(1.21)	(1.21)	(1.17)	(1.20)
Worked full year (50+ wks)	-12.35***	-13.01***	-10.33***	-16.85***	-8.29***	-16.89***
	(0.60)	(1.20)	(0.59)	(1.11)	(0.69)	(0.82)
Number of children	-0.46**	0.03	-0.10	-0.92**	-0.41	-0.52
	(0.23)	(0.39)	(0.22)	(0.42)	(0.26)	(0.33)
Child under 6	1.30**	-2.22**	0.97*	-0.35	-0.07	-1.28
	(0.59)	(0.97)	(0.52)	(1.19)	(0.59)	(1.46)
New baby this year	5.67***	5.22**	4.69***	6.51**	5.54***	13.64
	(1.49)	(2.17)	(1.30)	(2.73)	(1.25)	(9.34)
Observations	22,114	6,786	20,648	8,252	15,384	13,516

Notes: Linear probability models of leaving teaching (percentage points) estimated separately by subsample, with year effects, ASEC weights, calendar 2015–2024; heteroskedasticity-robust standard errors in parentheses. "–" marks regressors without variation in the subsample.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Table 7. Paid Family Leave and the Birth-Year Exit

	Leaves teaching
New baby	6.98***
	(0.82)
New baby × paid family leave	−2.87*
	(1.54)
Paid family leave active	0.04
	(0.30)
State and year effects	Yes
Demographic controls	Yes
Observations	77,883
Births under active programs	268

Notes: Weighted linear probability model of leaving teaching (percentage points) on an indicator for a new baby, interacted with an indicator for the state having an active paid family leave program in the teaching year. First full benefit years: California 2005, New Jersey 2010, Rhode Island 2014, New York 2018, Washington 2020, District of Columbia and Massachusetts 2021, Connecticut 2022, Oregon 2024. Standard errors, clustered by state, in parentheses.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.